

How To:

Import NMRtist results
into
CcpNmr Analysis version 3



www.nmrtist.org

```
data_nef_file_from_cyana

# NMR meta data
save_nef_nmr_meta_data
  nef_nmr_meta_data.sf_category      nef_nmr_meta_data
  nef_nmr_meta_data.sf_framecode    nef_nmr_meta_data
  nef_nmr_meta_data.format_name     nmr_exchange_format
  nef_nmr_meta_data.format_version  1.1
  nef_nmr_meta_data.program_name    CYANA
  nef_nmr_meta_data.program_version 4.0
  nef_nmr_meta_data.creation_date   2024-11-26T23:48:55
  nef_nmr_meta_data.uuid            CYANA-2024-11-26T23:48:55-5385537942

  loop_
    nef_program_script.program_name
    nef_program_script.script_name
    nef_program_script.script

REMARK Contents:4
;
nef_molecular_system      151 residues
nef_nmr_spectrum_C13HSQC_hmqc_@FLYA_asn  707 peaks with 450 assignments
nef_nmr_spectrum_C13NOESY_13cnoesyarom_@FLYA_asn  3948 peaks with 213 assignments
nef_nmr_spectrum_CBCANH_cbcanh_@FLYA_asn  801 peaks with 208 assignments
nef_nmr_spectrum_CBCAcoNH_cbcacoh_@FLYA_asn  427 peaks with 108 assignments
nef_nmr_spectrum_CcoNH_ccconh_@FLYA_asn  622 peaks with 151 assignments
nef_nmr_spectrum_HBHAcOH_hbhacoh_@FLYA_asn  598 peaks with 60 assignments
nef_nmr_spectrum_HCCHTOCSY_hcchtocsy_@FLYA_asn  1530 peaks with 359 assignments
nef_nmr_spectrum_HNcaCO_hncaco_@FLYA_asn  433 peaks with 0 assignments
nef_nmr_spectrum_N1SHSQC_hsqc_@FLYA_asn  255 peaks with 129 assignments
nef_nmr_spectrum_N1SNOESY_noesyhsqc_@FLYA_asn  6416 peaks with 918 assignments
;
stop_

save_

# Molecular system
save_nef_molecular_system
  nef_molecular_system.sf_category      nef_molecular_system
  nef_molecular_system.sf_framecode    nef_molecular_system

  loop_
    nef_sequence.index
    nef_sequence.chain_code
    nef_sequence.sequence_code
    nef_sequence.residue_name
    nef_sequence.linking
    nef_sequence.residue_variant

1 . 1 SER start .
2 . 2 MET middle .
3 . 3 VAL middle .
4 . 4 SER middle .
5 . 5 PRO middle .
6 . 6 GLU middle .
7 . 7 GLU middle .
8 . 8 ALA middle .
9 . 9 VAL middle .
```

Chemical Shift Table	
ChemicalShiftList CL:shifts	
Value (ppm)	NmrAtom
5.648	NA:A.1.HIS.HA
4.028	NA:A.1.HIS.HBx
4.369	NA:A.1.HIS.HBy
8.662	NA:A.1.HIS.HD1
7.400	NA:A.1.HIS.HD2
7.955	NA:A.1.HIS.HE1
56.113	NA:A.2.MET.CA
31.486	NA:A.2.MET.CB
27.054	NA:A.2.MET.CE
31.501	NA:A.2.MET.CG
8.996	NA:A.2.MET.H
4.118	NA:A.2.MET.HA
3.102	NA:A.2.MET.HBx
3.240	NA:A.2.MET.HBy
3.102	NA:A.2.MET.HGx
3.239	NA:A.2.MET.HGy
113.996	NA:A.2.MET.N
55.577	NA:A.3.ARG.CA
31.252	NA:A.3.ARG.CB
43.227	NA:A.3.ARG.CD
27.544	NA:A.3.ARG.CG
8.448	NA:A.3.ARG.H
4.376	NA:A.3.ARG.HA
1.615	NA:A.3.ARG.HBx
1.844	NA:A.3.ARG.HBy
3.134	NA:A.3.ARG.HD%
1.615	NA:A.3.ARG.HG%
121.984	NA:A.3.ARG.N
53.722	NA:A.4.GLN.CA
29.259	NA:A.4.GLN.CB

Introduction

This tutorial is designed to give an overview of the steps involved in importing the results from NMRtist peak picking and chemical shift assignments into a CcpNmr Analysis V3 project. We assume that you are already familiar with the basics of how the program works, e.g. by doing our Beginner Tutorial.

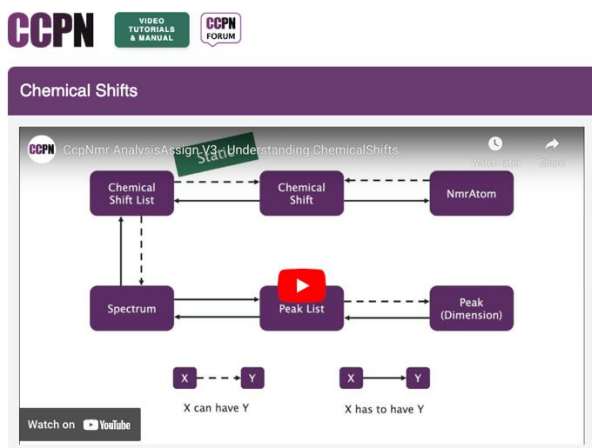
You can download the example data to go with this tutorial from the tutorials page of our website (<https://ccpn.ac.uk/support/tutorials/>).

This tutorial does not give any instructions about the NMRtist server itself; users should use the tutorials and guidelines available on the NMRtist website: <https://nmrtist.org/>

Additionally, we recommend users familiarize themselves with a few concepts from the CcpNmr Analysis manual:

1. Handling chemical shifts in relation to spectra:

<https://www.ccpn.ac.uk/manual/v3/ChemicalShifts.html>



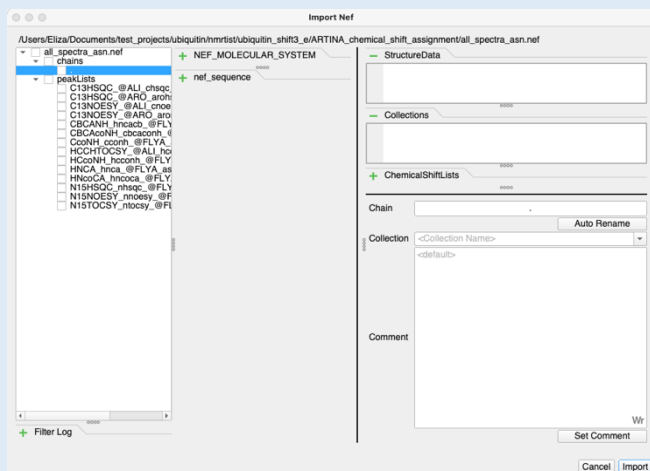
2. Basic information about the NEF format

<https://www.ccpn.ac.uk/manual/v3/NEF.html>

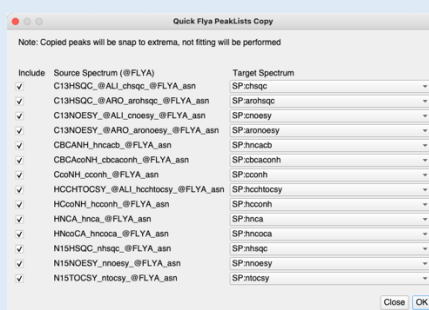
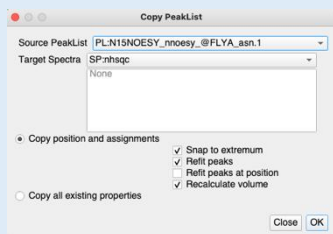
Note that images are representative, and that there may be small differences between your setup and that shown in the tutorial.

Main Steps

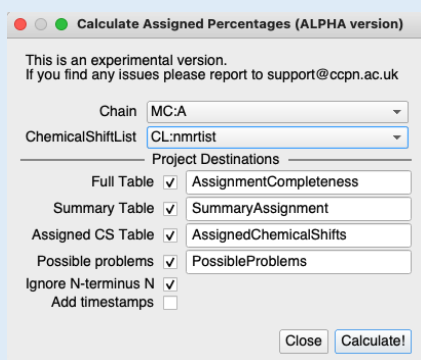
- Open project UbiquitinSpectraOnly.ccpn
- If using ARTINA structure determination: merge NEF files with CCPN Macro mergeMultiNef
- Import NMRtist Peak Lists from NEF file
 - Update Chain Code
 - Name Chemical Shift List for imported Peak Lists



- Copy Peak Lists to existing Spectra



- Inspect Peak Lists and Chemical Shifts

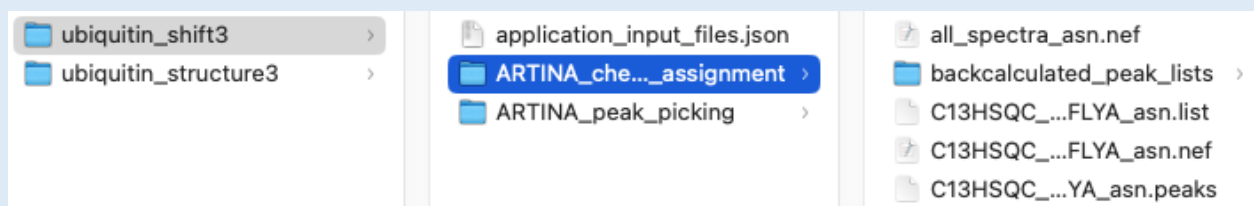


Different NMRtist Applications will produce NEF files with slightly different content.

In this document we will focus on importing peak lists, which contain assigned peaks. These will automatically generate a chemical shift list.

ARTINA: chemical shift assignment

This application gives two folders (May 2025):

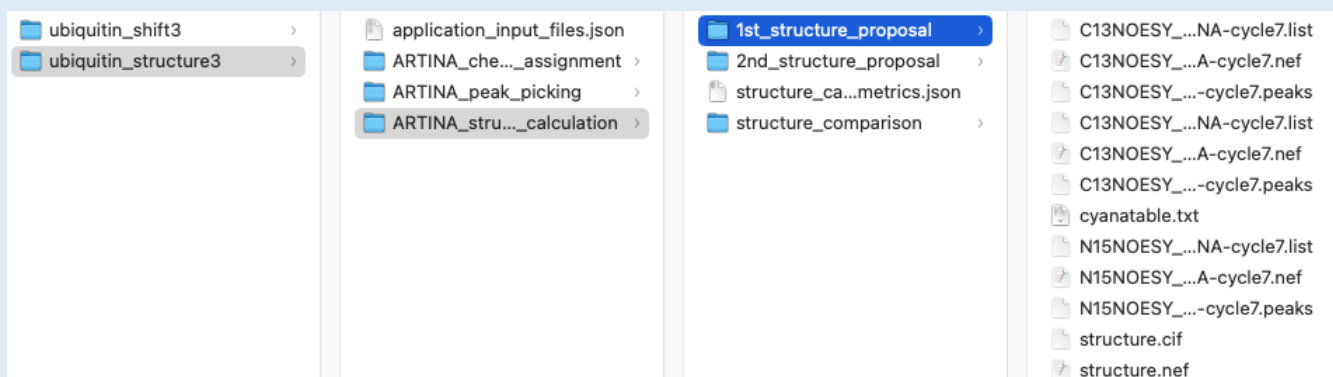


The **ARTINA_chemical_shift_assignment** folder contains:

- **all_spectra_asn.nef** - contains all peak lists (spectra), but no chemical shift list
- **shifts.nef** & **shifts-strong.nef** - contain chemical shift lists, but no peak lists
- NEF files containing individual peak lists (spectra)
- **backcalculated_peak_lists** folder - NEF files with individual peak lists (spectra)

ARTINA: structure determination

This application gives three folders:



The **ARTINA_structure_calculation/1st_structure_proposal** (or **2nd_structure_proposal**) folder contains:

- **structure.nef** – contains a chemical shift list and distance restraints (these can differ between the 1st and 2nd proposals), but no peak lists.

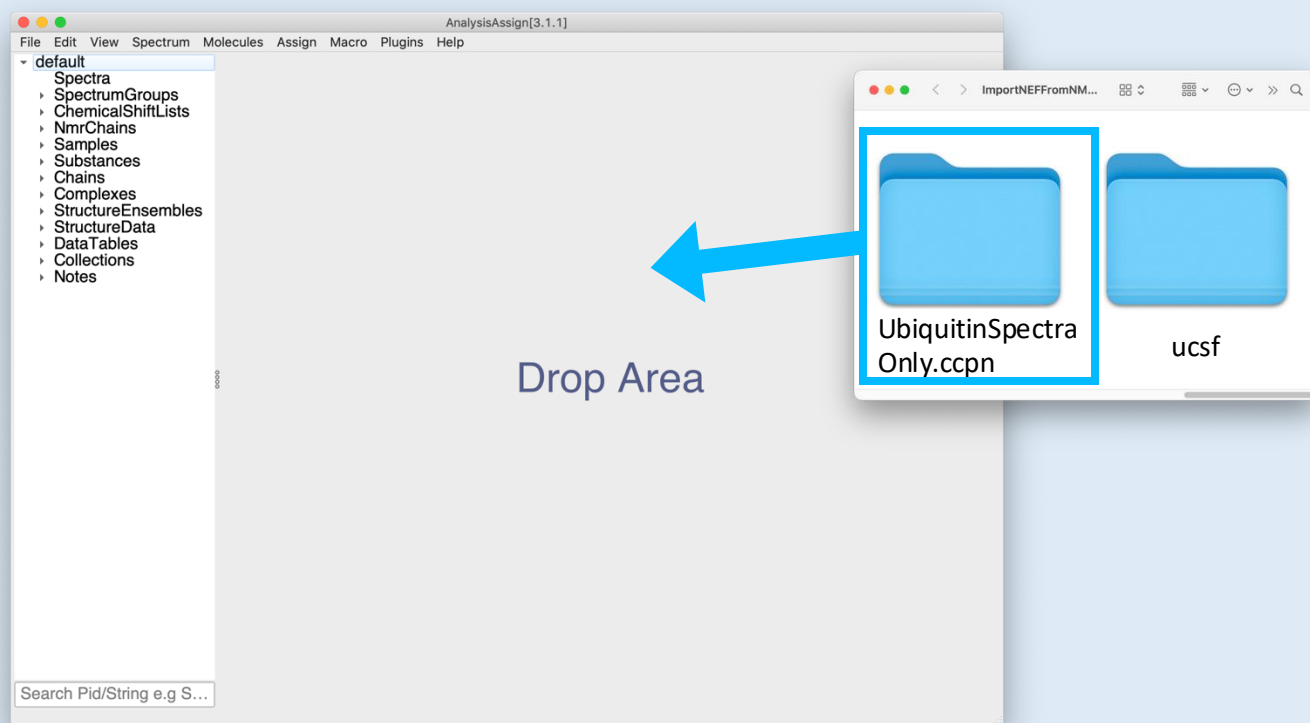
The **ARTINA_chemical_shift_assignment** folder contains:

- NEF files containing individual peak lists (spectra)

Note: unlike above, there is no NEF file containing all peak lists (spectra)!

Note: there are small differences between the chemical shifts loaded from files such as **shifts.nef**, **structure.nef** and **all_spectra_asn.nef**.

All Artina's NEF files contain sequence information.



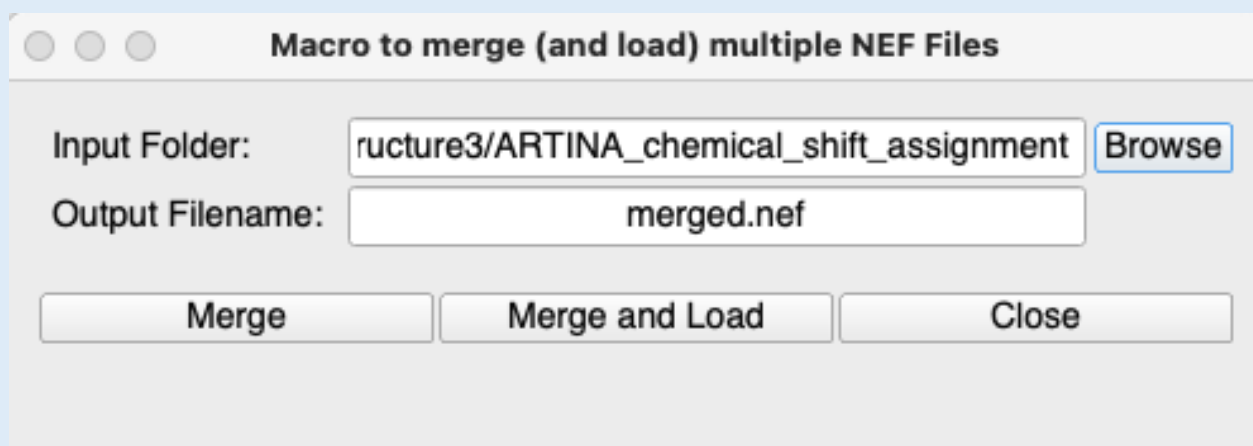
1A Load Project

CcpNmr projects are saved as directories with the **.ccpn** extension.

- Find the **UbiquitinSpectraOnly.ccpn** project in the **ImportNEFFFromNMRtist** tutorial data directory and drag & drop it into the program.

The project contains all the spectra (with Experiment Types already set up) which were used for an NMRtist run.

- ▾ UbiquitinSpectraOnly
 - ▾ Spectra
 - SP:nhsqc
 - SP:chsqc
 - SP:arohsqc
 - SP:hnca
 - SP:hncoca
 - SP:cbcaconh
 - SP:hncacb
 - SP:cconh
 - SP:hccconh
 - SP:ntocsy
 - SP:nnoesy
 - SP:hcchtocsy
 - SP:cnoesy
 - SP:aronoesy

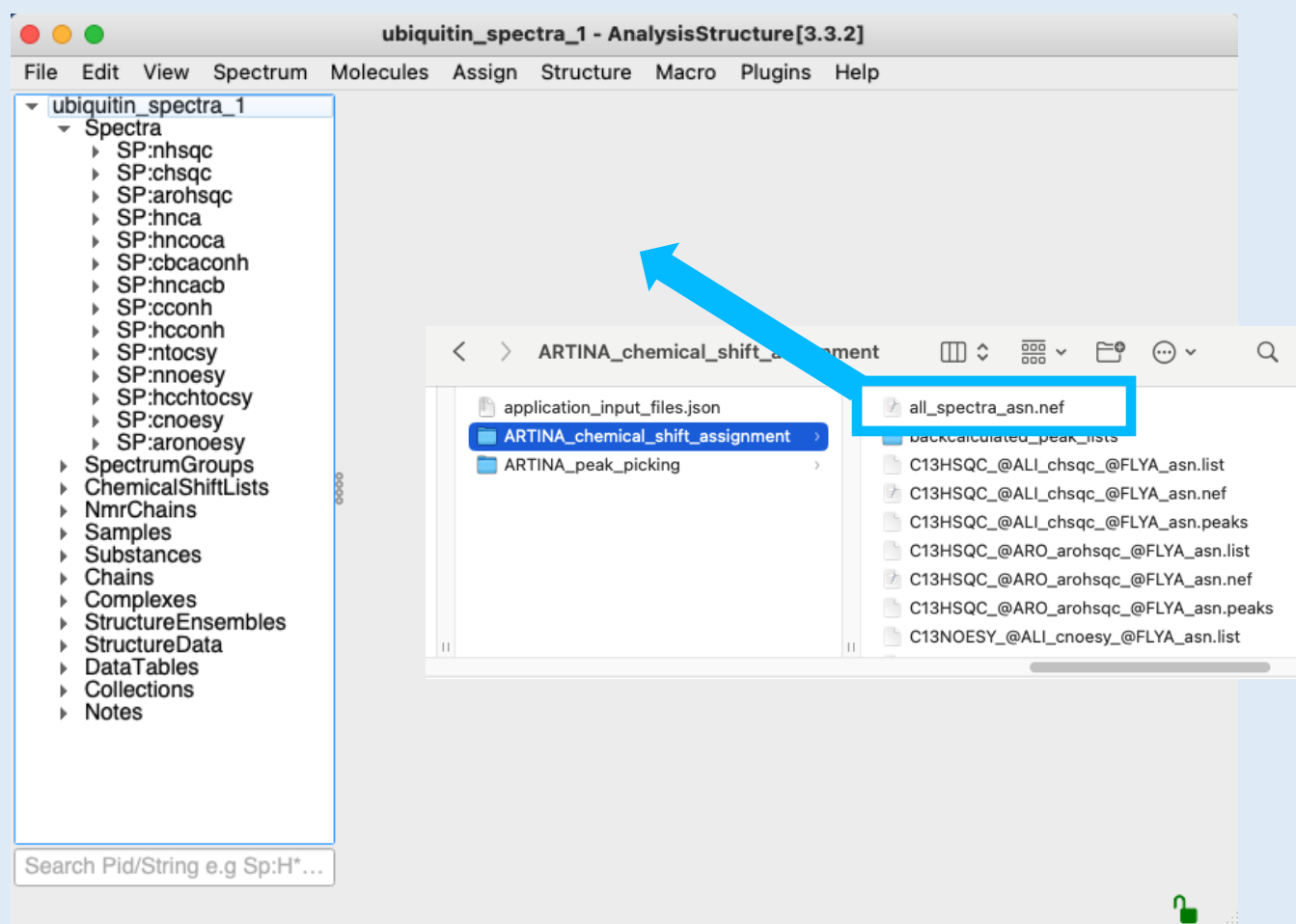


1B Accessing ARTINA structure determination NEF files

As mentioned on page 3, the **ARTINA structure determination** application will not generate a NEF file that includes all peak lists together. The **mergeMultiNef** CCPN Macro will create and load such a NEF file by merging the individual NEF files for each spectrum/peak list.

- Unzip the **ubiquitin_structure3.zip** file in the **nmrtist** subfolder of the tutorial folder (**ImportNEFFromNMRtist**). This zip file is the file obtained from the NMRtist server.
- Go to **Macro > Run CCPN Macros > mergeMultiNef**
- In the resulting popup type in the path to the folder containing the files you wish to merge or navigate to the folder using the **Browse** button.
For the tutorial data this is the **ImportNEFFromNMRtist / nmrtist / ubiquitin_structure3 / ARTINA_chemical_shift_assignment** folder.
- Update the name of the resulting file if you wish.
- Click **Merge and Load**, and in the resulting popup click **Import**
- *Alternatively:* Click **Merge** and load the NEF file as described in point **1C**
- Now continue with steps **1D – 1F**

Note: This macro can also be used to merge NEF files from the **ARTINA_chemical_shift_assignment / backcalculated_peak_lists** folder if desired.

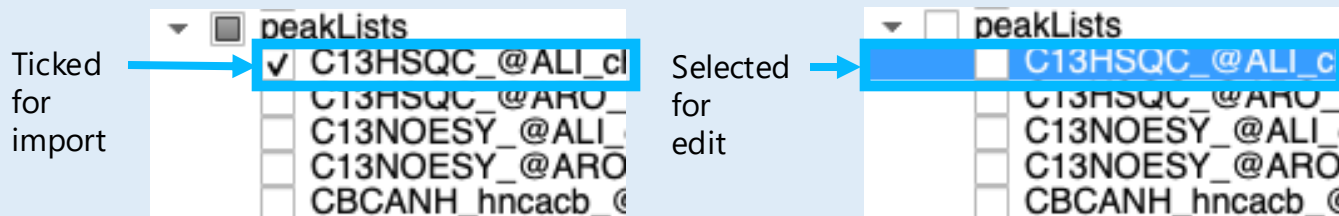
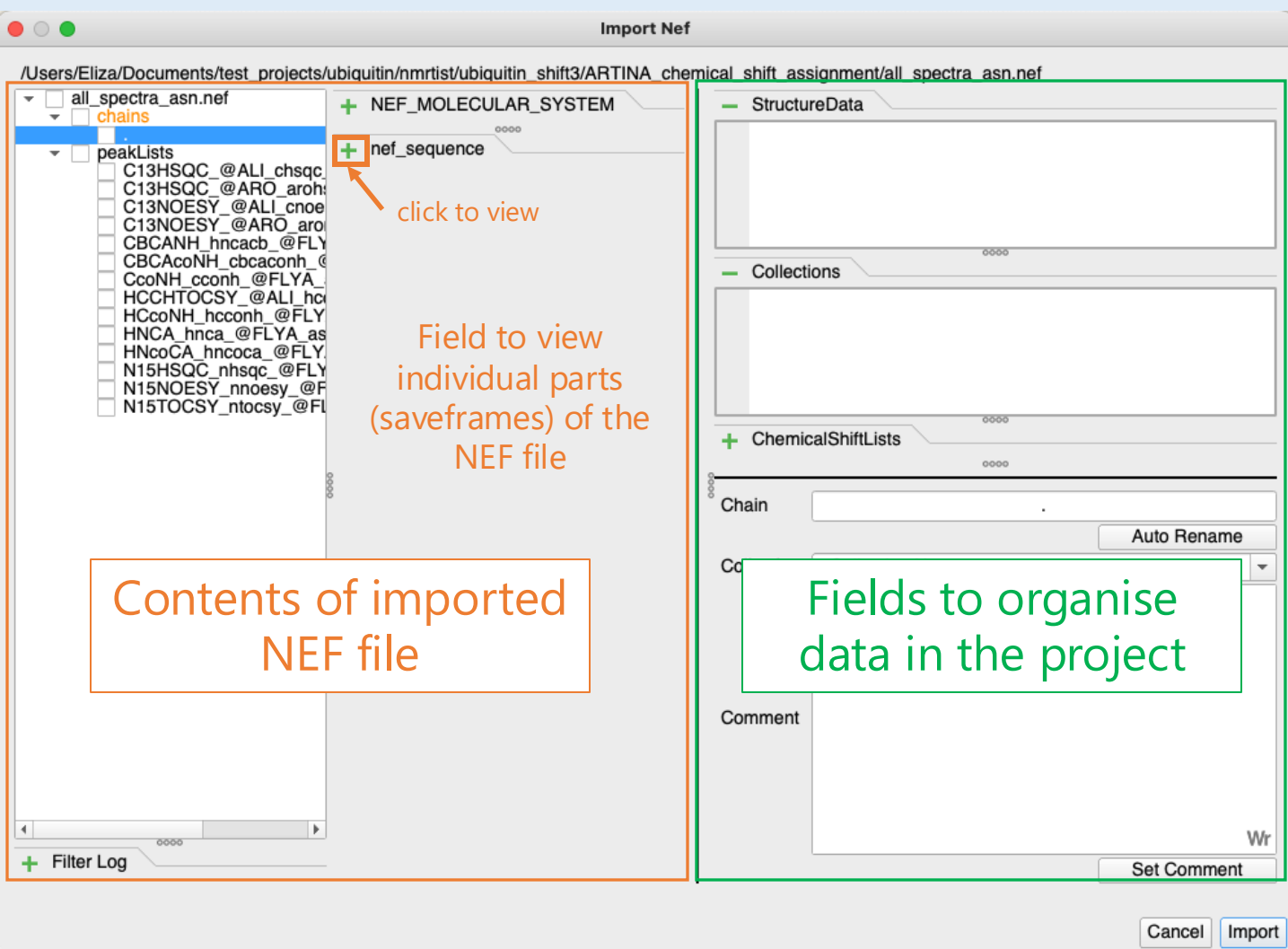


1C Accessing ARTINA chemical shift assignment NEF files

- Unzip the **ubiquitin_shift3.zip** file in the **nmrlist** subfolder of the tutorial folder (**ImportNEFFromNMRlist**). This zip file is the file obtained from the NMRlist server.
- Go to the **ARTINA_chemical_shift_assignment** folder.
- Go to **File / Import > Nef File** and select the **all_spectra_asn.nef** file.

OR

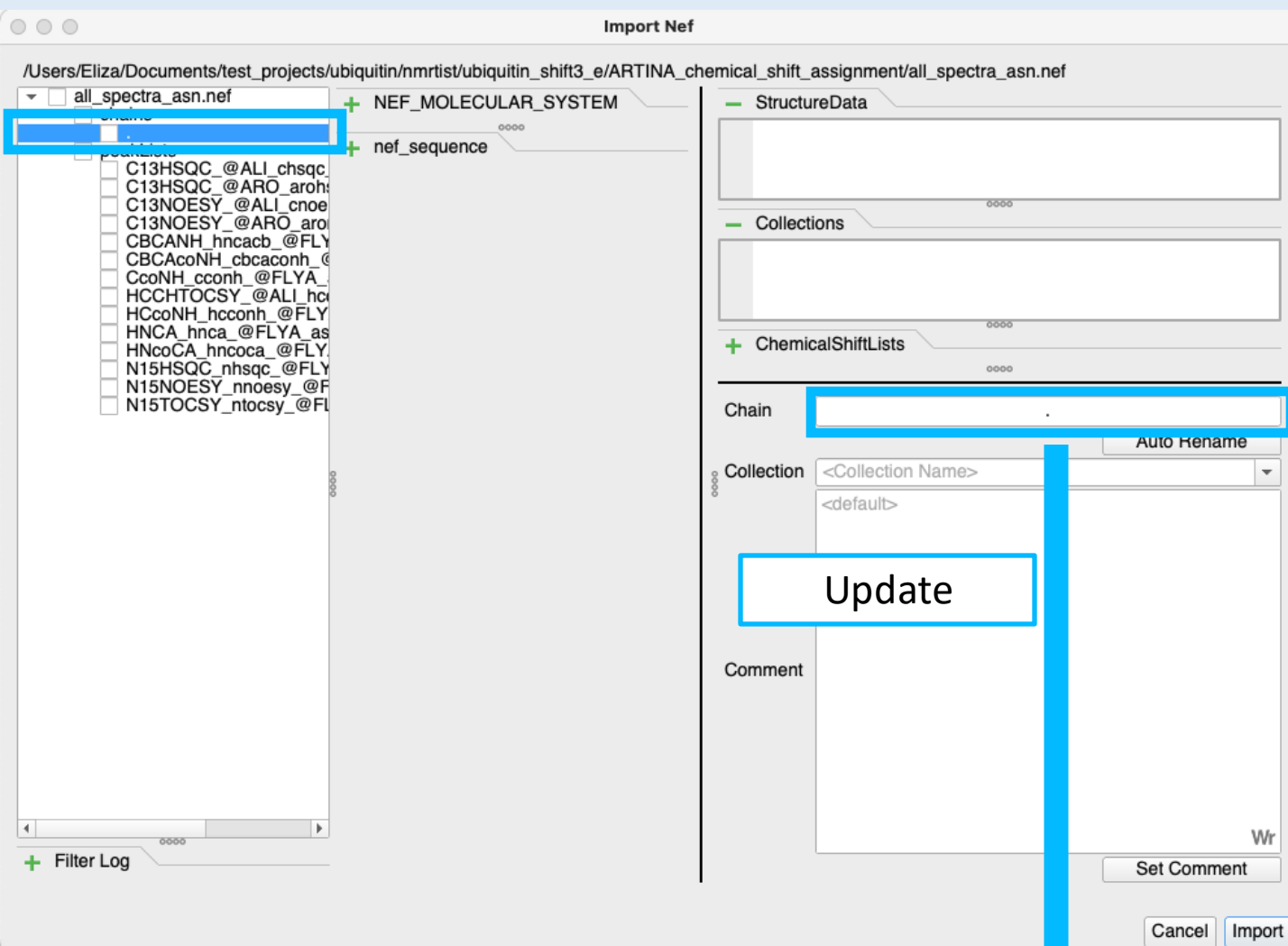
- Drag & Drop the **all_spectra_asn.nef** file into the Drop Area of the program.
- At the prompt click **Import**.
- You will see the Import NEF pop up window opening.



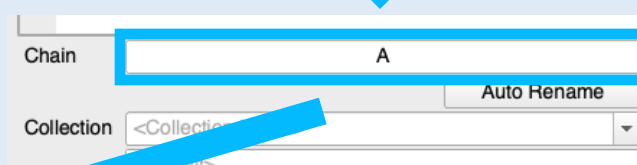
1D The NEF Importer popup

- Items chosen for import must be ticked
- You can select item(s) for editing by clicking on the item (the background will be highlighted in blue). If you start to edit an item, it will be ticked by default.
- You can view/inspect the contents of items by clicking on the green plus sign in the middle of the pop-up window.

NMRtist provides a NEF file in which no symbol is used for the Chain Code. We recommend changing it to a capital letter, as is common practice elsewhere.



NEF_MOLECULAR_SYSTEM				
nef_sequence				
	index	chain_code	residue	residue_offset
1	1	A	1	1
2	2	A	2	2
3	3	A	3	3
4	4	A	4	4



Press 'Return'

1E Loading and updating the Chain from the NEF file

- Tick '.', this will highlight the chain section ready for editing.
- On the right hand side in the **Chain** field change the name to '**A**' (or another name of your choice). **Press 'Return' to confirm.**
- If you expand the section on the left side of the window you can see the Chain Code is now updated.

NEF files imported from the **ARTINA_chemical_shift_assignment** folder do not have a chemical shift list associated with the peak lists. This should be updated during import.

The screenshot shows the 'Import Nef' dialog box. On the left, the file tree shows 'all_spectra_asn.nef' with 'chains' and 'peakLists' expanded. On the right, the 'ChemicalShiftLists' section has two dropdowns: 'collection' and 'chemicalshiftlist'. A blue box highlights the 'chemicalshiftlist' dropdown, and a blue arrow points to it with the text 'Update'. Below this, another blue box highlights the 'chemicalshiftlist' dropdown with the value 'nmrtist' selected, and a blue arrow points to it with the text 'Press \'Return\''. At the bottom, a table shows the 'NEF_NMR_SPECTRUM_C13HSQC_@ALI_CHSQC_@FLY' with columns 'attribute' and 'value'. The 'chemical_shift_list' attribute is highlighted with the value 'nef_chemical_shift_list_nmrtist'.

attribute	value
1 sf_category	nef_nmr_spectrum
2 sf_framecode	nef_nmr_spectrum_C13HSQC...
3 num_dimensions	2
4 chemical_shift_list	nef_chemical_shift_list_nmrtist
5 experiment_type	C13HSQC
6 cyana_experiment_type	C13HSQC

1F Naming chemical shift list for the imported peak lists

- Click on **peakLists** on the left-hand side.
This will highlight all spectra ready for editing.
- On the right-hand side in the **ChemicalShiftList** field change the name to **nmrtist** (or another name of your choice).
- Press **Return** to confirm.
- If you expand the section on the left-hand side of the window for any of the spectra, you can see the chemical shift list name is now updated.
All PeakLists should be automatically ticked for import
- Click **Import**, which may take few minutes, depending on the number of spectra.

Peak lists will be imported as separate spectra associated with separate chemical shift list. The next steps is to copy all or selected peak lists to their target spectra.

Drag and drop

Copy PeakList

Source PeakList: PL:N15HSQC_nhsqc_@FLYA_asn.1

Target Spectra: > select-to-add <

SP:nhsqc

☒ Copy position and assignments

☒ Snap to extremum

☒ Refit peaks

☐ Refit peaks at position

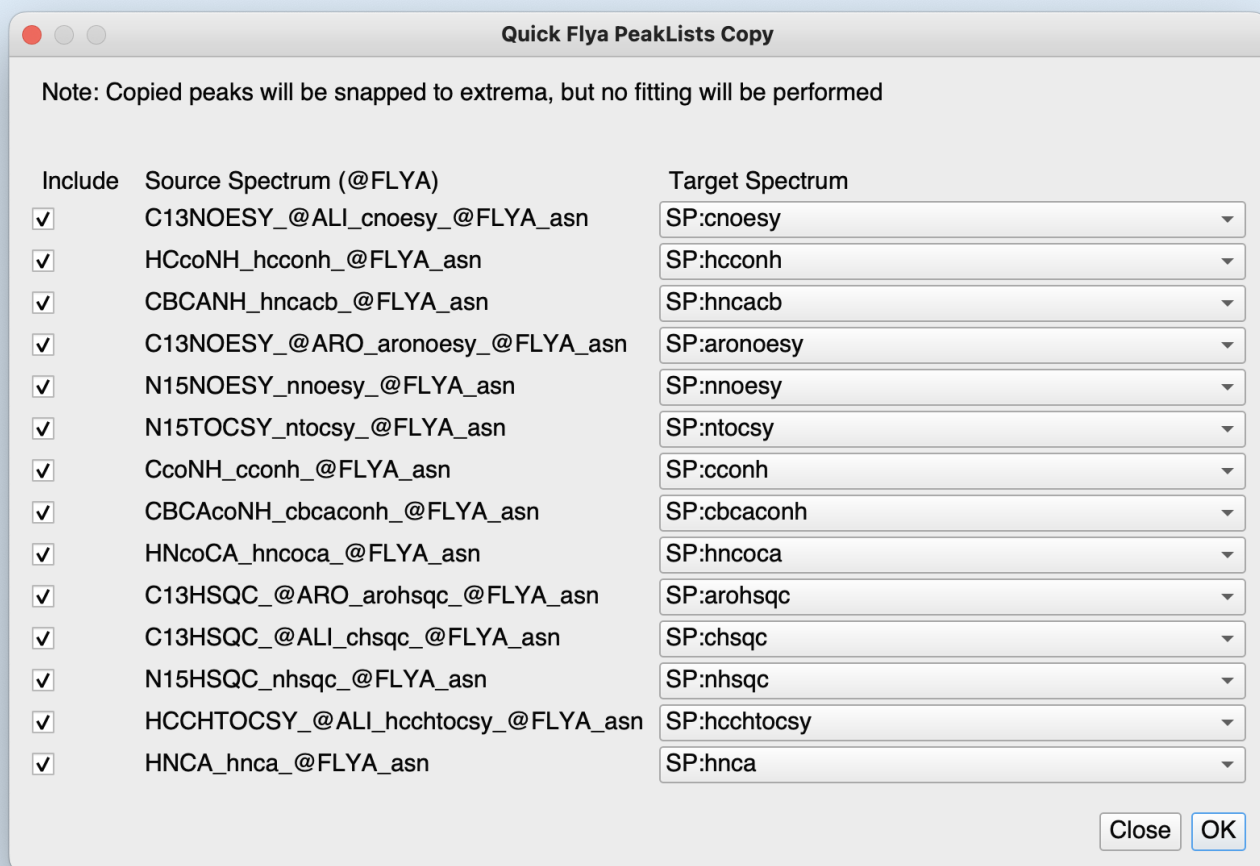
☒ Recalculate volume

☐ Copy all existing properties

Close OK

2A Copy peak list to the target spectra

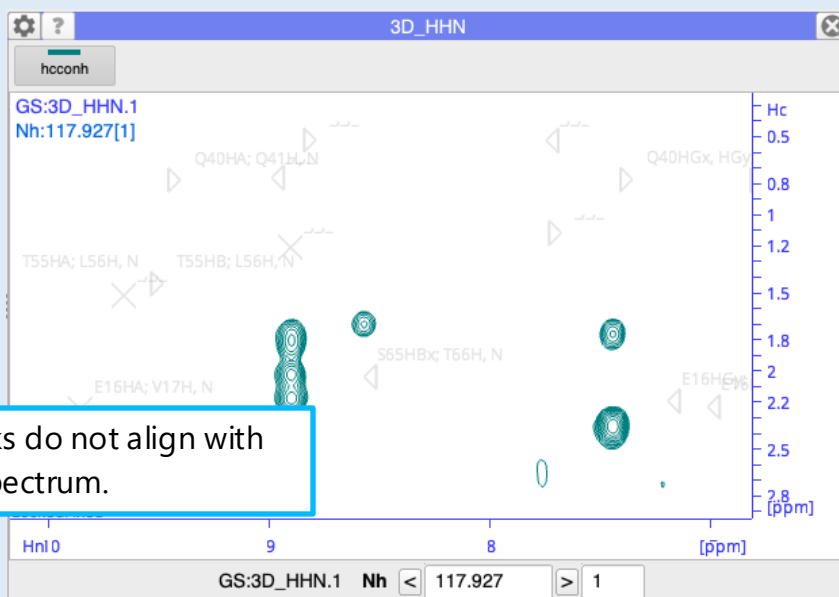
- Drop the **SP:nhsqc** spectrum into the DropArea to display it.
- Expand the **SP:N15HSQC_nhsqc_@FLYA_asn** spectrum in the sidebar to view its peak lists.
- Drag and drop the **PL:N15HSQC_nhsqc_@FLYA_asn.1** peak list onto the SpectrumDisplay. This will open the **Copy PeakList** popup.
- Make sure the **SP:nhsqc** spectrum is selected as the Target Spectrum.
- You can keep all the settings as they are or untick the **Recalculate volume** (as volumes are not needed and this adds extra time)
- *Alternatively:* Go to **Spectrum > Copy PeakLists...**



This macro automatically pairs spectra containing '@FLYA' in their names with matching target spectra in the project.

2B Copying multiple '@FLYA' peak lists to target spectra simultaneously

- Go to **Macro > Run CCPN Macros > quickFlyaCopy**
- Click **OK**
- Depending on the number of spectra it may take few minutes to run this macro.
- **Note:** only peak positions and assignments will be copied. If fitting is necessary, it needs to be performed separately on each peak list.
- *Known Issue:* you may get a message with a warning ("position constrain... violated by value..."), it **should be** harmless, we are working to resolve this.



Spectrum Properties: HCcoNH_hconh_@FLYA_asn

General Dimensions Contours

Dimension Axis Code

Axis Code	Nh	Hc	Hn
Isotope Code	15N	1H	1H
Coherence Order	SQ	SQ	SQ
Reference Experiment Type	HcoNH		
Reference Experiment Dimensions	Nh	Hc	Hn
Magnetisation Transfers	dimension1	dimension2	transferType isIndirect

Copy to Axis Codes

Spectrum Properties: HCCHTOCSY_@ALI_hcchtocsy_@FLYA_asn

General Dimensions Contours

Dimension Axis Code

Axis Code	Ch1	Hc1	Hc
Isotope Code	13C	1H	1H
Coherence Order	SQ	SQ	SQ
Reference Experiment Type	HCCH-TOCSY		
Reference Experiment Dimensions	Ch1	Hc1	Hc
Magnetisation Transfers	dimension1	dimension2	transferType isIndirect

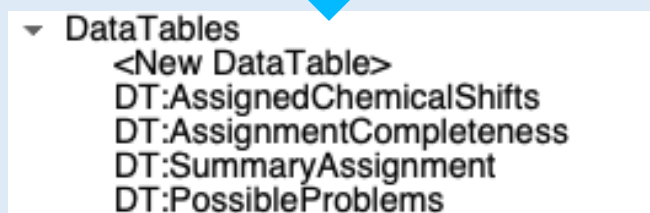
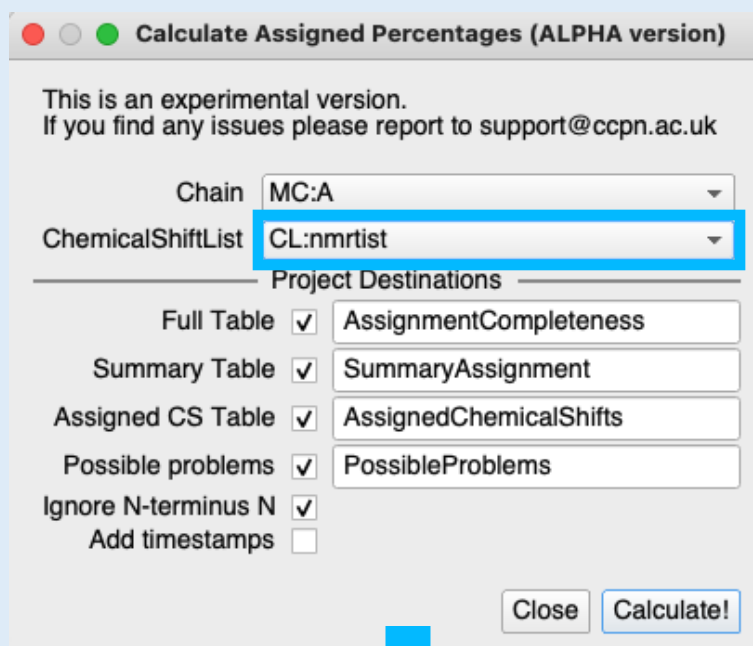
Copy to Axis Codes

3A Inspecting copied peaks

- Drag **SP:hconh** onto drop area to display it. You can see that the peaks were not copied to the correct dimensions.
- Open the **SP:HCcoNH_hconh_@FLYA_asn Spectrum Properties** popup (double-click on the spectrum in the sidebar)
- Set the **Reference Experiment Type** and **Reference Experiment Dimensions** as shown above and click on **Copy to Axis Codes**.

Note: the order of the spectrum dimensions in the imported spectra is different than it was in the original spectra!

- Copy the PeakList from **SP:HCcoNH_hconh_@FLYA_asn** to **SP:hconh** again
- The HCCH-Tocsy will require a similar correction.



?

Data Table

DataTable

DT:PossibleProblems

comment

Problematic Chemical Shift

+

Metadata

pid	residueType	sequenceCode	C	CA	CB	CD	CE	CZ	H	HA	HB	HD	HE	HZ	N
MR:A.4.PHE	PHE	4	0.0	1.0	1.0	1.0	0.5	0.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0

This is very good quality data with a large number of spectra, consequently there will only be a very small number of errors (though there are some!).

3B Inspecting chemical shift list

- Go to **Macro > Run CCPN Macros > ProteinAssignmentStatistics**
- Select the **nmrtist** chemical shift list to be inspected, click **Calculate!**

Four new DataTables will be created.

- Drag **DT:PossibleProblems** into the Drop Area to display it.

You can see that **PHE.4** was flagged with an issue as only half the **CE** carbons are reported. The **PHE.4** chemical shifts and assignments should be inspected in detail.

- Note:** It is common for Artina to assign aromatic ring atoms with separate shifts (non-degenerate, e.g. **HD1** and **HD2**), in most cases, where the shift difference is very small, those should be annotated as degenerate (**HD%**).

Contact Us

Website:

www.ccpn.ac.uk

Suggestions and comments:

support@ccpn.ac.uk

Issues and bug reports:

<https://www.ccpn.ac.uk/forums/>

Cite Us

Skinner, S. P. *et al.* CcpNmr AnalysisAssign: a flexible platform for integrated NMR analysis. *J. Biomol. NMR* (2016). doi:10.1007/s10858-016-0060-y

Cite NEF

Gutmanas, A. *et al.* NMR Exchange Format: A unified and open standard for representation of NMR restraint data. *Nature Structural and Molecular Biology* 22, 433–434 (2015).